Introduction to DNA METHYLATION and METHYLOME

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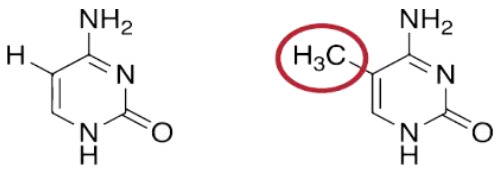
Outline

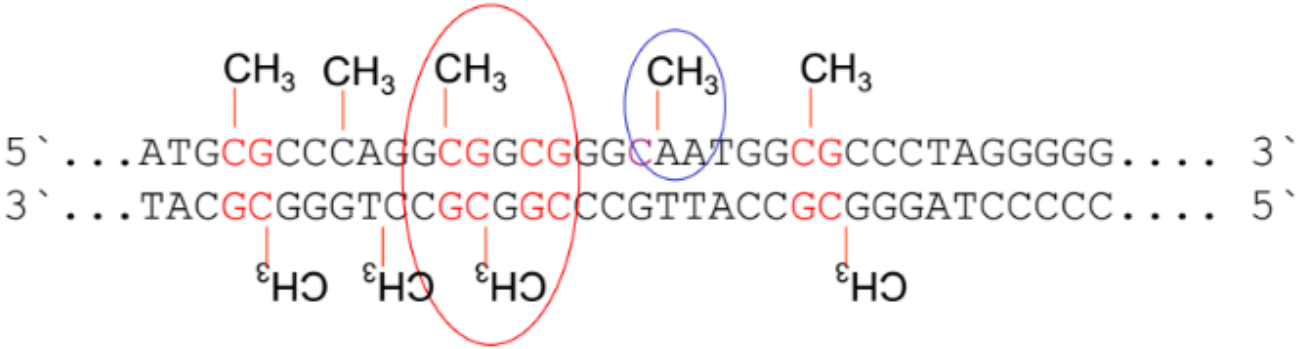
- Introduction to
- DNA methylation
- Profiling DNA methylation
- Whole Genome Bisulphite Sequencing (WGBS)
- Differential Methylated Region (DMR)
- IGV to visualize DNA methylation
- Enhancer and Promoter DNA methylation

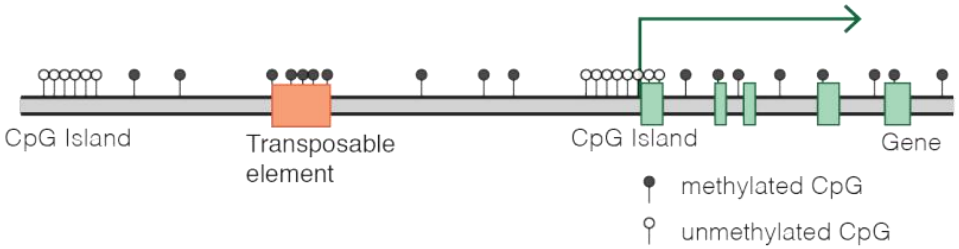
What are DNA Methylation and genome-wide DNA methylation (Epigenome)?

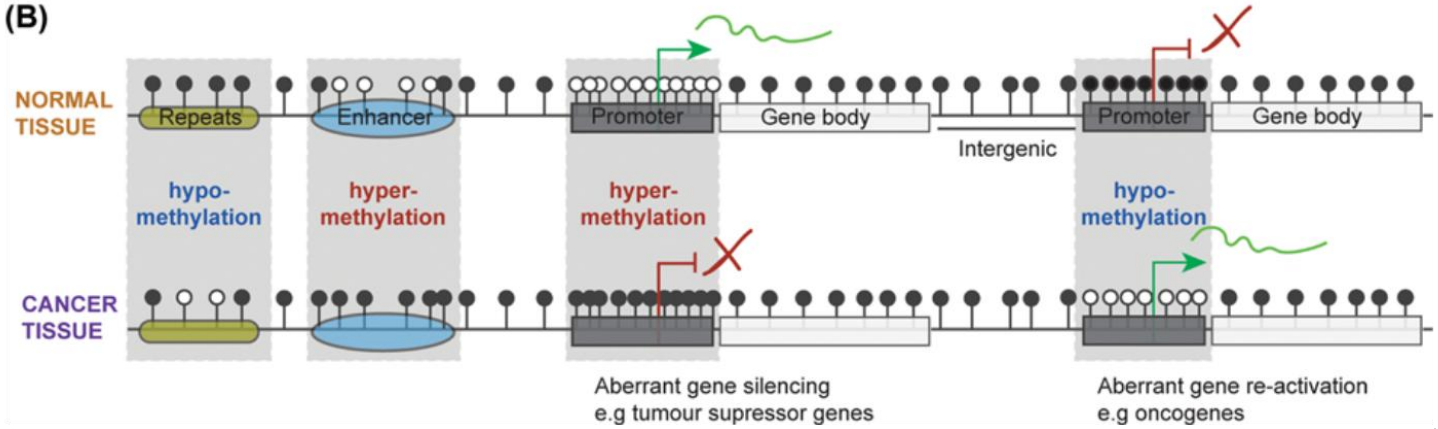


Cytosine **methylated Cytosine**

Typical mammalian DNA methylation landscape



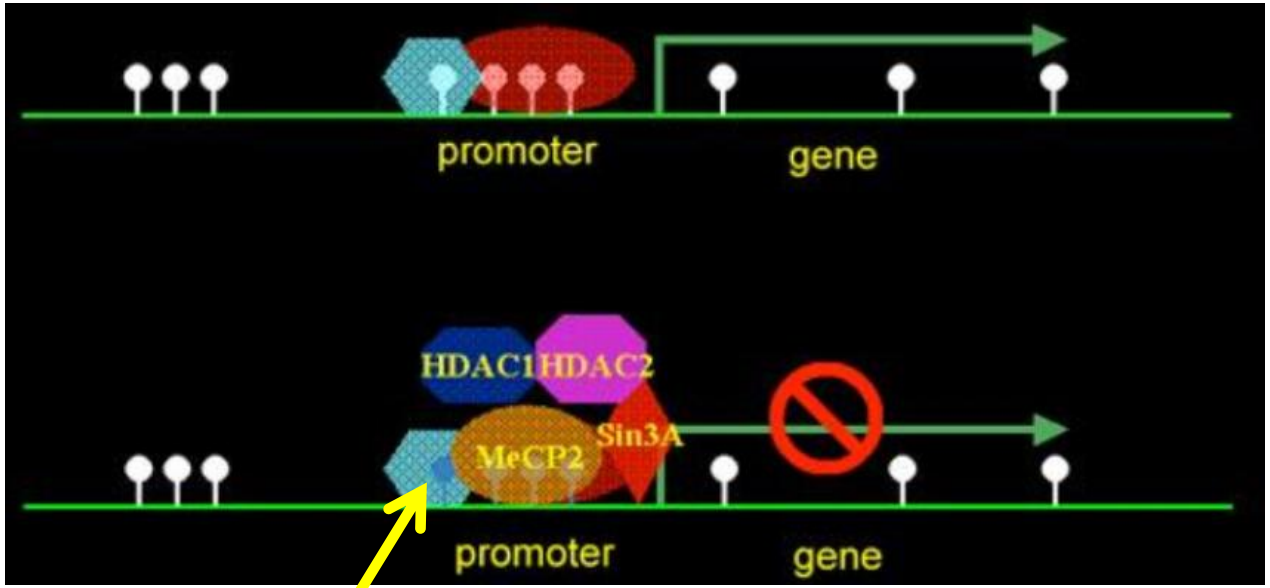




https://en.wikipedia.org/wiki/DNA_methylation

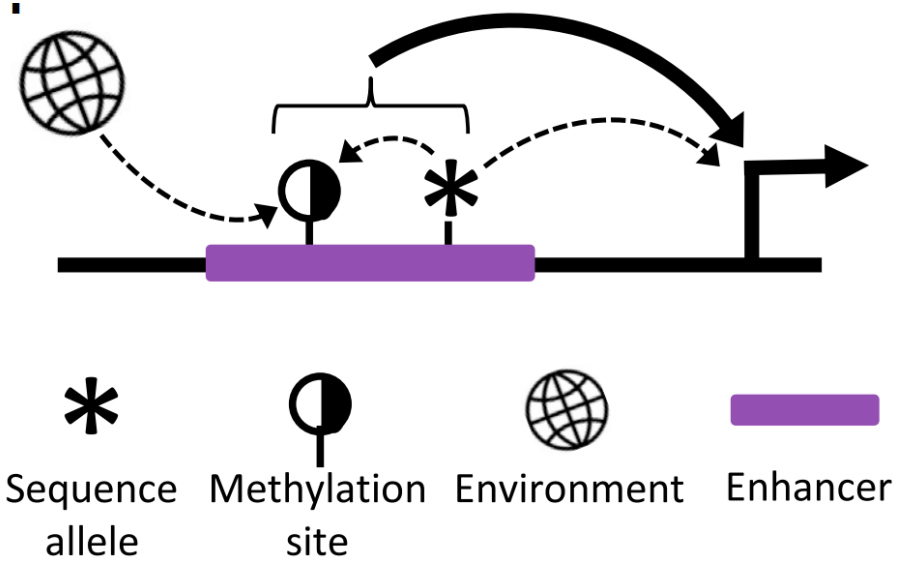
DNA methylation regulates gene expression

Promoter



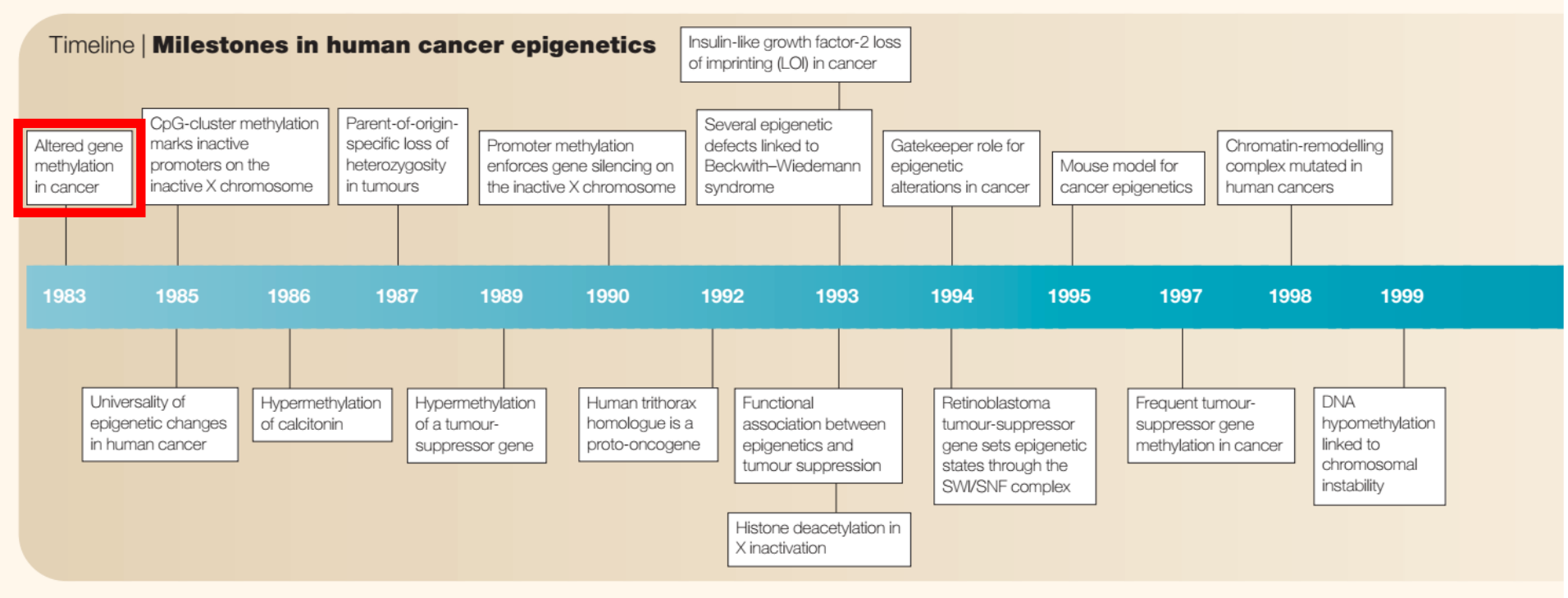
Methylated CpG site

Enhancer



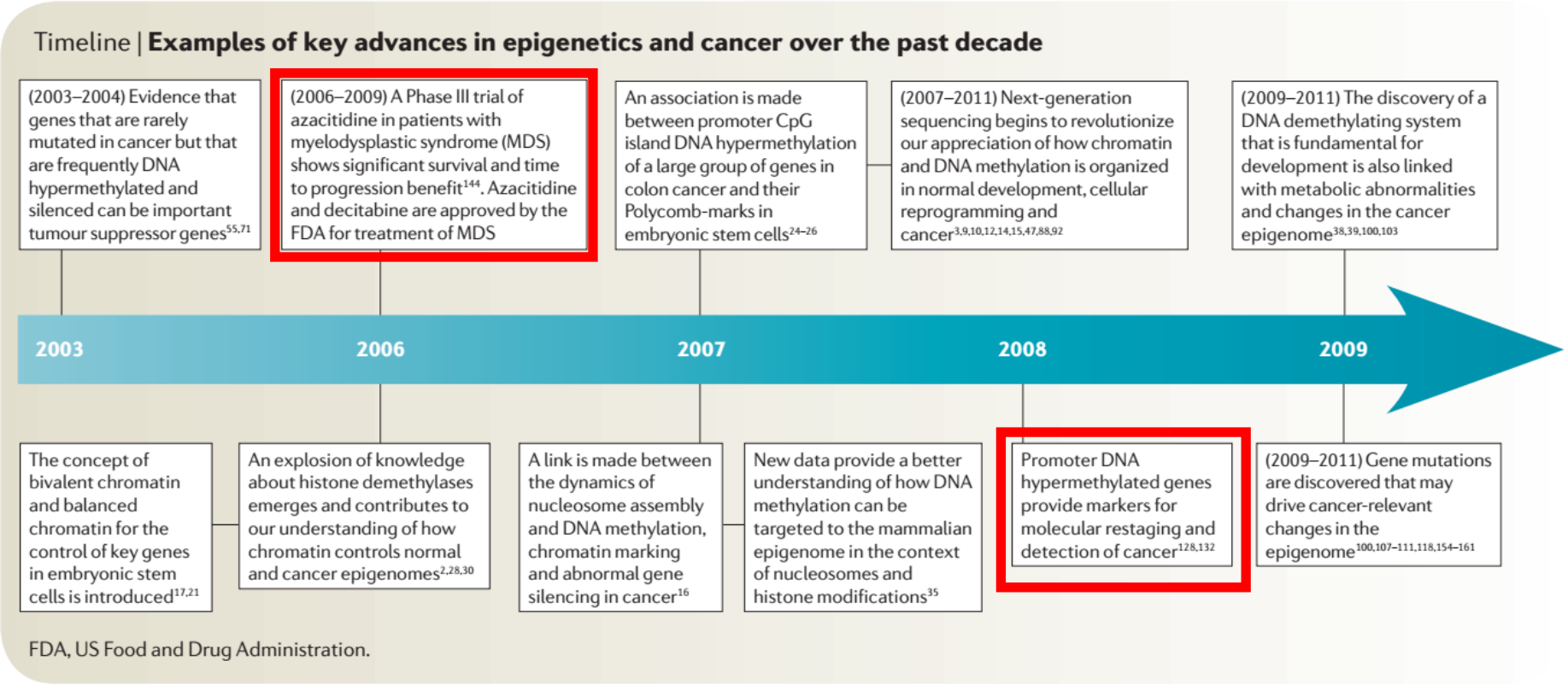
10.1016/j.cell.2013.06.018

DNA methylation regulate gene expression in **cancer**



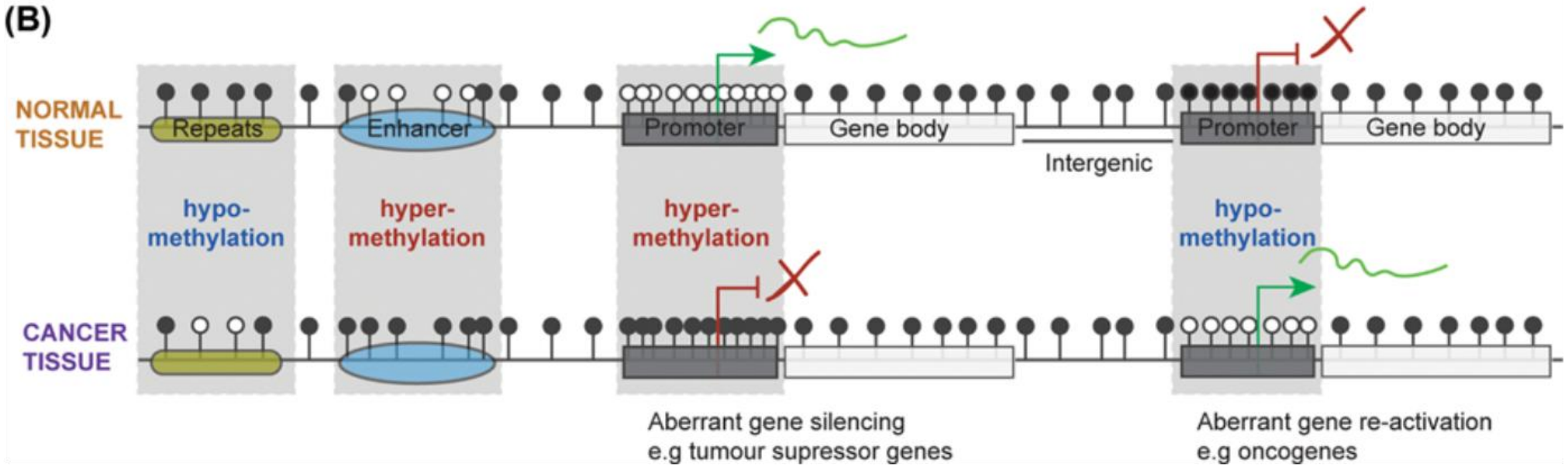
10.1038/nrc1279

DNA methylation regulate gene expression in **cancer**

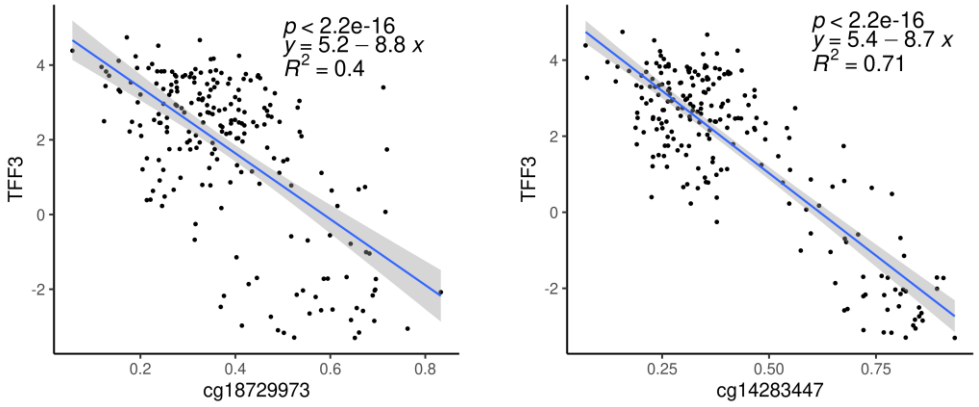


10.1038/nrc3130

DNA methylation regulate gene expression in cancer



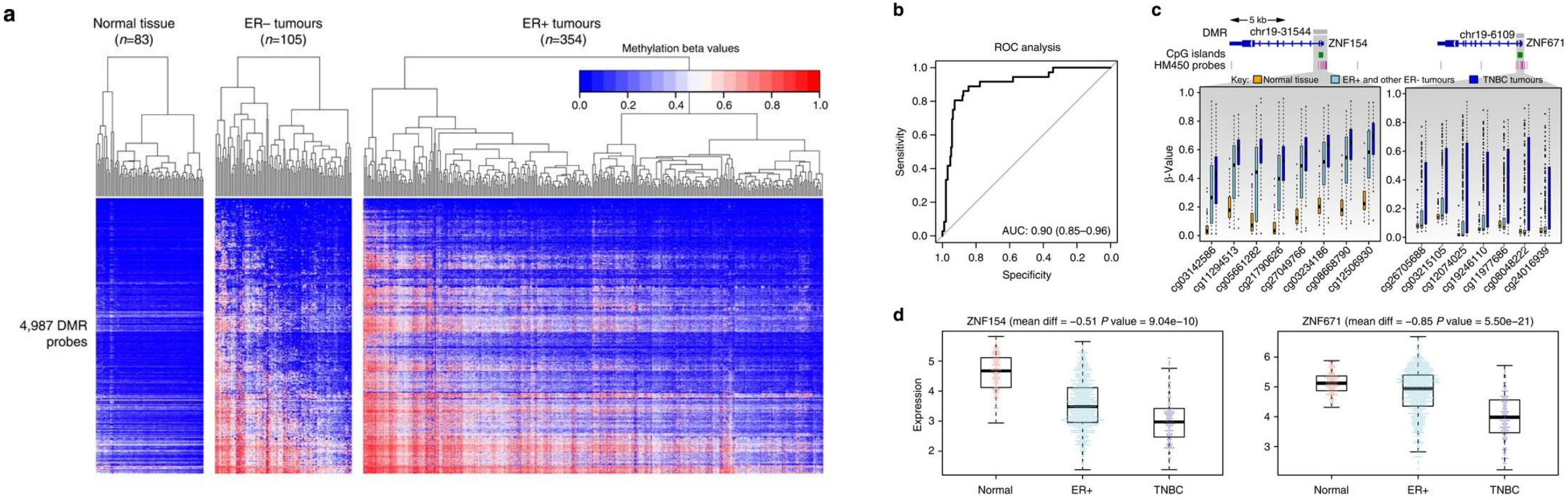
Expression



DNA methylation

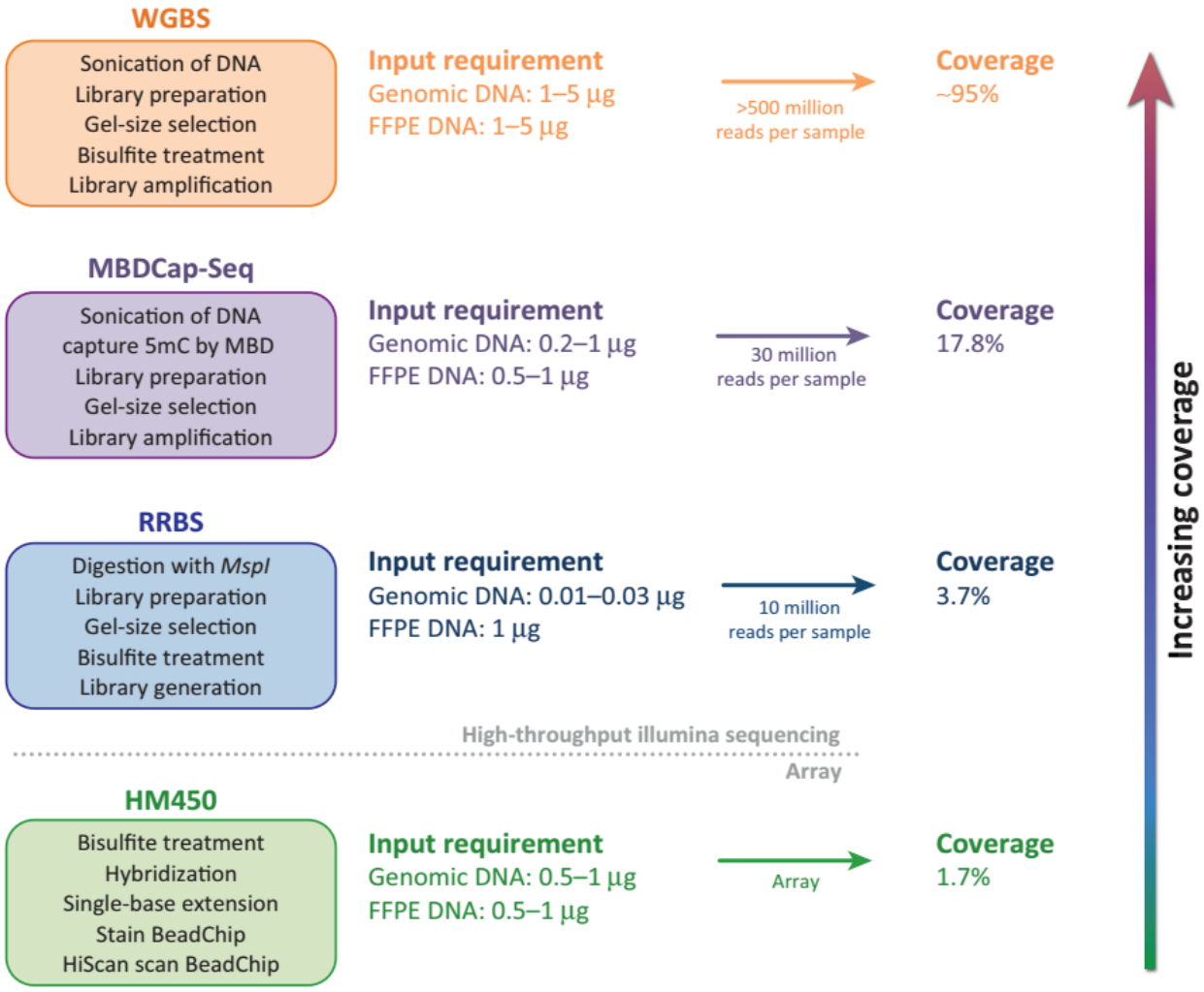
10.1042/EBC20190037

DNA methylation regulate gene expression in breast cancer



Stirzaker C., Zotenko E., Song J.Z., Qu W., Nair S.S., Locke W.J., Stone A., Armstong N.J., Robinson M.D., Dobrovic A... **SJ. Clark**. Methylome sequencing in triple-negative breast cancer reveals distinct methylation clusters with prognostic value. Nat. Commun. 2015

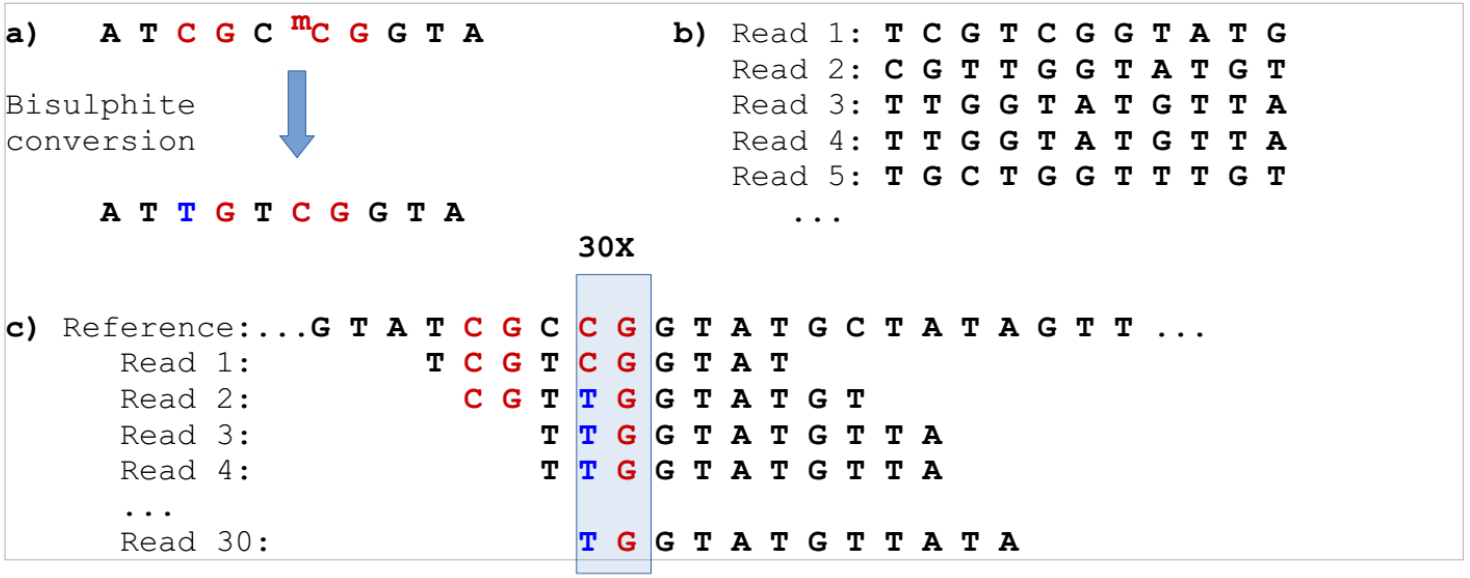
Methods to profile DNA methylation



850K/EPIC

10.1016/j.tig.2013.11.004

Bisulphite Sequencing



Beta value = Percent of Methylation (Perc)

Perc = 100*Meth/Cov

⇒ Meth = Perc * Cov /100

⇒ Cov = C + T

In the example:

Perc = 100*1/5 = 20%

Meth = 1 = 20 * 5/100

Cov = 1 + 4 = 5

***Figure 1**. a) Sodium bisuphlite converts unmethylated cytosines to thymines. b) The converted DNA is sequenced. c) Alignment and aggregation of these short sequence can be used to quantify the ratio of methylated (C) to unmethylated (converted T) cytosines from a population of cells at a single locus.*

Total_cov_matrix_CpG.txt

	A	B	C	D
1	chr	pos	nmDNA1	nmDNA2
2	chr2	133031228	56	55
3	chr2	133031240	74	81
4	chr2	133031244	82	79
5	chr2	133031249	64	70
6	chr2	133031265	65	69
7	chr2	133031268	67	56

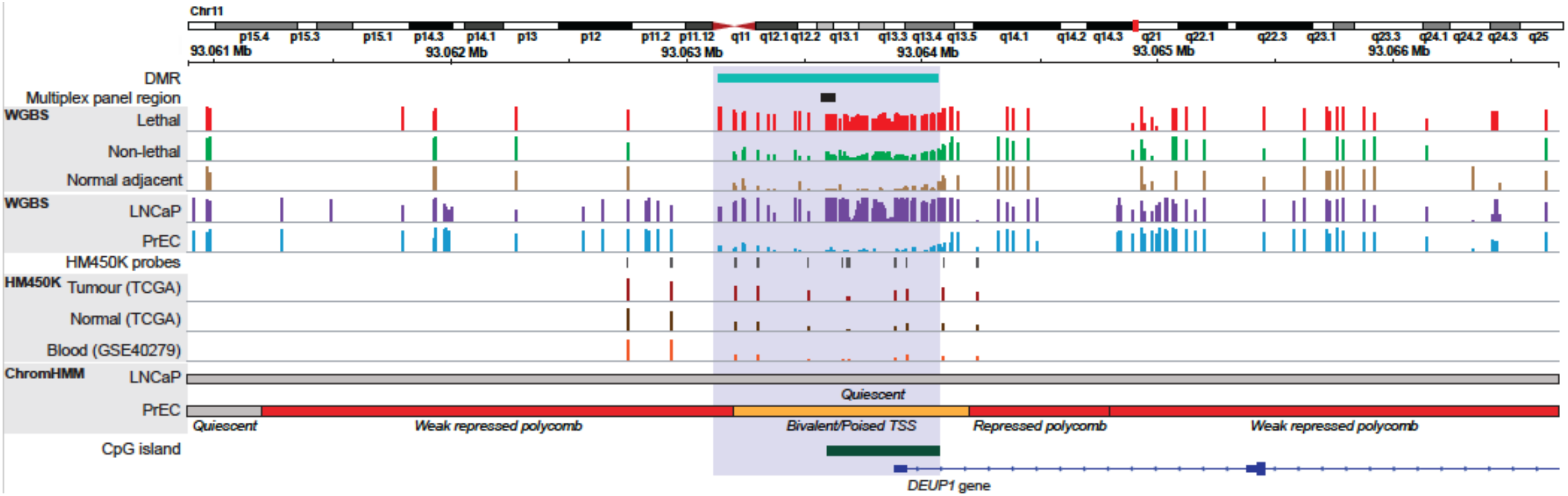
Meth_cov_matrix_CpG.txt

	A	B	C	D
1	chr	pos	nmDNA1	nmDNA2
2	chr2	133031228	41	28
3	chr2	133031240	29	19
4	chr2	133031244	47	47
5	chr2	133031249	35	24
6	chr2	133031265	19	21
7	chr2	133031268	24	16

Perc_cov_matrix_CpG.txt

	A	B	C	D
1	chr	pos	nmDNA1	nmDNA2
2	chr2	133031228	73	51
3	chr2	133031240	39	23
4	chr2	133031244	57	59
5	chr2	133031249	55	34
6	chr2	133031265	29	30
7	chr2	133031268	36	29

WGBS vs 450K


Chr11
p15.4 p15.3 p15.1 p14.3 p14.1 p13 p12 p11.2 p11.12 q11 q12.1 q12.2 q13.1 q13.3 q13.4 q13.5 q14.1 q14.2 q14.3 q21 q22.1 q22.3 q23.1 q23.3 q24.1 q24.2 q24.3 q25
93.061 Mb
93.062 Mb
93.063 Mb
93.064 Mb
93.065 Mb
93.066 Mb
DMR
Multiplex panel region
WGBS
Lethal
Non-lethal
Normal adjacent
WGBS
LNCaP
PrEC
HM450K probes
HM450K
Tumour (TCGA)
Normal (TCGA)
Blood (GSE40279)
ChromHMM
LNCaP
Quiescent
PrEC
Quiescent
Weak repressed polycomb
Bivalent/Poised TSS
Repressed polycomb
Weak repressed polycomb
CpG island
DEUP1 gene

Quiz: Genetics vs Epigenetics

- Cancer: Hypermethylation of promoter regions (e.g., BRCA1, MLH1)
- PWS/Angelman: Imprinting errors (deletion or silencing of chromosome 15q11-q13)
- Type 2 Diabetes: Altered methylation in pancreatic cells and insulin-sensitive tissues
- Autoimmune (SLE): Global hypomethylation of DNA in immune T-cells

THANK YOU FOR YOUR ATTENTION!

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For further information!